

## Types of Test Metrics:

**Defect metrics**: Measure the number, severity, and density of defects found in the software.

**Defect Density**: The number of defects found per unit of code (e.g., per thousand lines of code).

**Defect Leakage**: The number of defects that were not caught during testing and were released to the end-user.

**Test coverage metrics**: Measure how much of the software's code or functionality has been tested.

**Test Coverage**: The percentage of total code that is executed by a test suite.

**Test Execution Coverage**: The percentage of planned test cases that have been executed.

**Test effort metrics**: Focus on the resources, time, and cost associated with testing.

**Test Execution Time**: The total time taken to execute test cases.

**Cost Per Test Case/Cost Per Defect**: The cost incurred to run a single test case or to fix a single defect.

## Test Estimation:

Test estimation is the process of predicting the time, effort, and resources needed for software testing activities. It is a crucial project management activity that helps in planning schedules, budgeting, and allocating resources effectively. Effective estimation involves breaking down tasks, considering factors like project complexity and staff skills, and continuously refining estimates throughout the project lifecycle.

What test estimation includes

**Effort:** The total hours (or other units like story points) required for testing tasks.

**Time:** The duration for completing testing, broken down into working days or calendar time, considering deadlines.

**Cost:** The budget needed for testing, which includes costs for resources, tools, and infrastructure.

### **Key steps for effective estimation:**

**Work Breakdown:** Decompose the entire testing process into smaller, manageable tasks, activities, and levels.

**Task Identification:** Define specific activities within each phase, such as test planning, analysis, design, and execution.

**Estimation:** Assign effort, time, and cost estimates to each individual task and then aggregate them to get an overall estimate.

**Factor Analysis:** Consider various factors that influence the estimate, including the complexity of the product, development model, and team competence and experience.

**Buffer Inclusion:** Add contingency time to account for unexpected issues or changes.

**Continuous Review:** Regularly compare your estimates against actual results from previous projects to improve accuracy for future work.

**Communication:** Clearly communicate your assumptions, the basis of your estimates, and potential trade-offs to stakeholders.